GUESS for Infrastructure and Transportation

Networks on maps, hubs and bridges, robustness to failure and attack, routes by hops and by distance, regional systems, with two worked case studies

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Why networks?

Airlines, railways, roads, power grids, pipelines and the internet are networks you can point to on a map. What the network view adds is how the parts depend on each other:

- Which places are hubs, and which are the bridges whose loss would cut regions off?
- How does the system degrade when parts fail at random, or are attacked?
- What is the best route by number of hops, and by distance?
- What regional systems does the network break into?

This book assumes the Kitchen for loading, querying and drawing. Its data are in the cookbooks' `data` folder:

file	what it is
`us-airports.gdf`	549 airports in the United States and the 2,787 airport pairs with scheduled domestic service. `weight` is the number of airline routes listed between the two. Node fields: `iata` (code), `city`, `latitude`, `longitude`, `airport` (name).
`world-airports.gdf`	The same for the whole world: 3,257 airports and 18,930 pairs, with `country`.
`power-grid-region.gdf`	A connected region of 400 of the 4,941 stations of the Western US power grid (Watts and Strogatz), laid out by Kamada–Kawai; positions are not geographic.
`internet-as-sample.gdf`	A snowball sample of 300 autonomous systems (the networks that make up the Internet) from a 2006 snapshot, with the links among them.

Both come from the OpenFlights airport and route databases (Open Database License), a 2014 snapshot. They record which routes exist, not how often they're flown or how many seats they carry, so a twice-weekly island hop counts the same as a shuttle.

Networks on maps

Put a network on a map Basic

You have nodes with latitude and longitude. **You want** them where they are, on a map.

Ingredients: `us-airports.gdf` · `geoLayout`

```
makeFromGDF("us-airports.gdf")
geoLayout(projection="albers")
sizeBy(degree, 3, 30)
colorBy(degree, lightgray, red)
hubs = sortBy(degree)[-12:]
hubs.labelvisible = true
g.edges.color = "150,150,150,40"
```

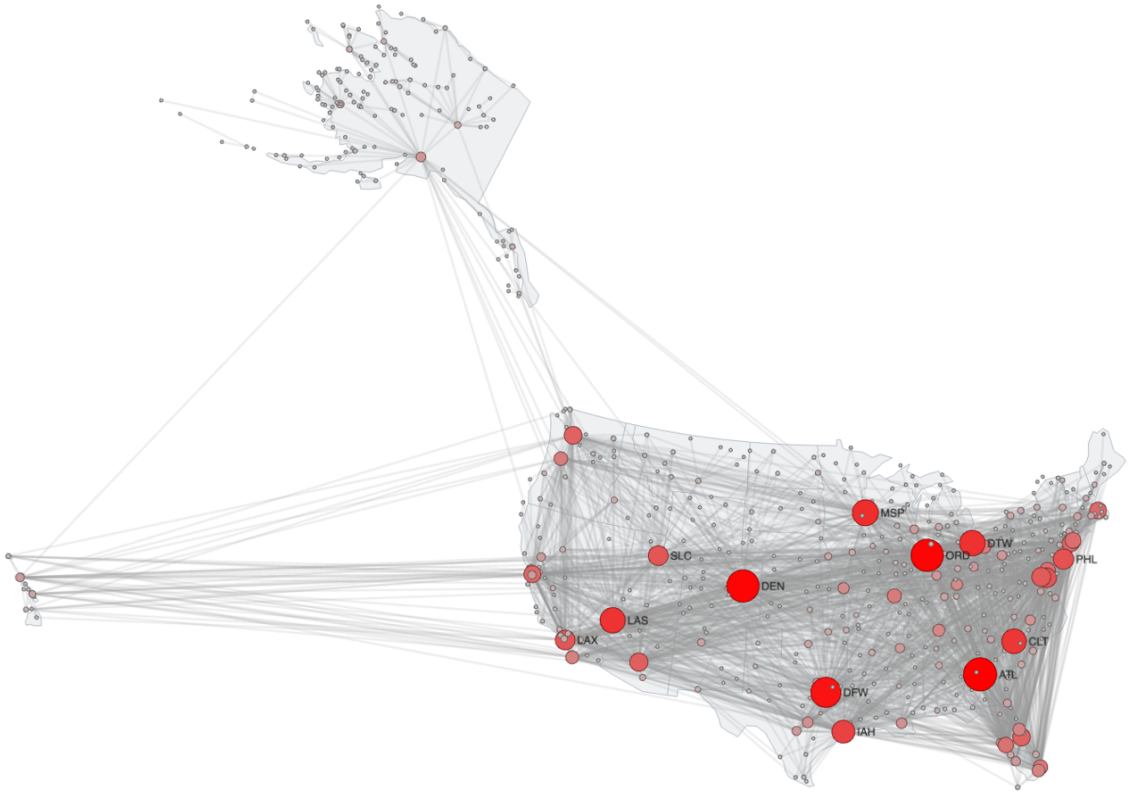


Figure 1: US domestic air routes on an Albers projection

How it works. `geoLayout()` finds the `latitude` and `longitude` fields (or `lat`/`lon`; name them if yours differ), projects each node, and draws a map underneath: the states when every node is in the United States, the world's coastlines otherwise. Its choice of projection matters:

projection	keeps	use for
`"mercator"` (default)	local shapes and directions	world maps, web-map habits
`"albers"`	areas; shapes good in mid-latitudes	countries and continents; the US by default (`parallels=`, `center=` for others)

projection	keeps	use for
`"equirectangular"`	nothing in particular, but simple	quick looks, data already in degrees

The map is drawn as vector outlines (Natural Earth, public domain). It stays sharp at any zoom, follows the light or dark theme (or `land=`/`borders=` colours), and is included in SVG and PDF exports. `removeMap()` removes it.

Pitfalls. The Albers projection tuned for the lower 48 states puts Alaska and Hawaii where they really are, so the map is wide and mostly ocean. Published US maps move them into insets. If that matters, open the lower 48 as a graph of its own, `subgraph((longitude > -130) & (latitude < 50))`, then run `geoLayout` again. Alaska gets the same treatment in the first case study.

A world network Basic

```
makeFromGDF("world-airports.gdf")
geoLayout()
sizeBy(degree, 2, 24)
colorBy(betweenness, lightgray, red, scale="log")
g.edges.visible = false
```

How it works. With 3,257 airports, the routes would cover the map, so this recipe hides them and lets size (number of routes) and colour (how often an airport lies on the shortest connections between others) carry the story. The busiest airports by routes are European: Amsterdam, Frankfurt, Paris and Istanbul, then Atlanta. By betweenness the list changes to Paris, Los Angeles, Anchorage, Dubai and Frankfurt. These are the connectors between world regions.

Take the network to Google Earth or a GIS Basic

You want the airports and routes in Google Earth, QGIS or ArcGIS, with their data.

Ingredients: `us-airports.gdf` · `exportKML` · `exportSHP` · `geoVectors`

```
makeFromGDF("us-airports.gdf")
colorBy(betweenness, lightgray, red)
exportKML("us-airports.kml")
exportSHP("us-airports.zip")
geoVectors()
```

How it works. Both exports take positions from the `latitude` and `longitude` fields, not from the layout, so they're right whatever the picture looks like. `exportKML` writes a placemark per airport and a line per route, in their GUESS colours, with every field in the placemark's data: open it in Google Earth. `exportSHP` writes a zip of two shapefiles (points for airports, lines for routes) in WGS 84, with the fields as attribute columns: drop it on QGIS or ArcGIS. `geoVectors()` stores each route's length in `km` and its `bearing`, and prints the mean route length.

Pitfalls. Shapefile column names are cut to 10 characters, so long field names come out shortened (and numbered when two would clash).

Colour a map Intermediate

You want a network of regions that share a border, for questions of adjacency: how many colours a map needs, which regions are neighbours, how far apart two regions are in border crossings.

Ingredients: `makeFromMap` · `colorGraph` · `edgeCrossings`

```
makeFromMap("usa")
colorGraph()
labelBy(abbrev)
edgeCrossings()
```

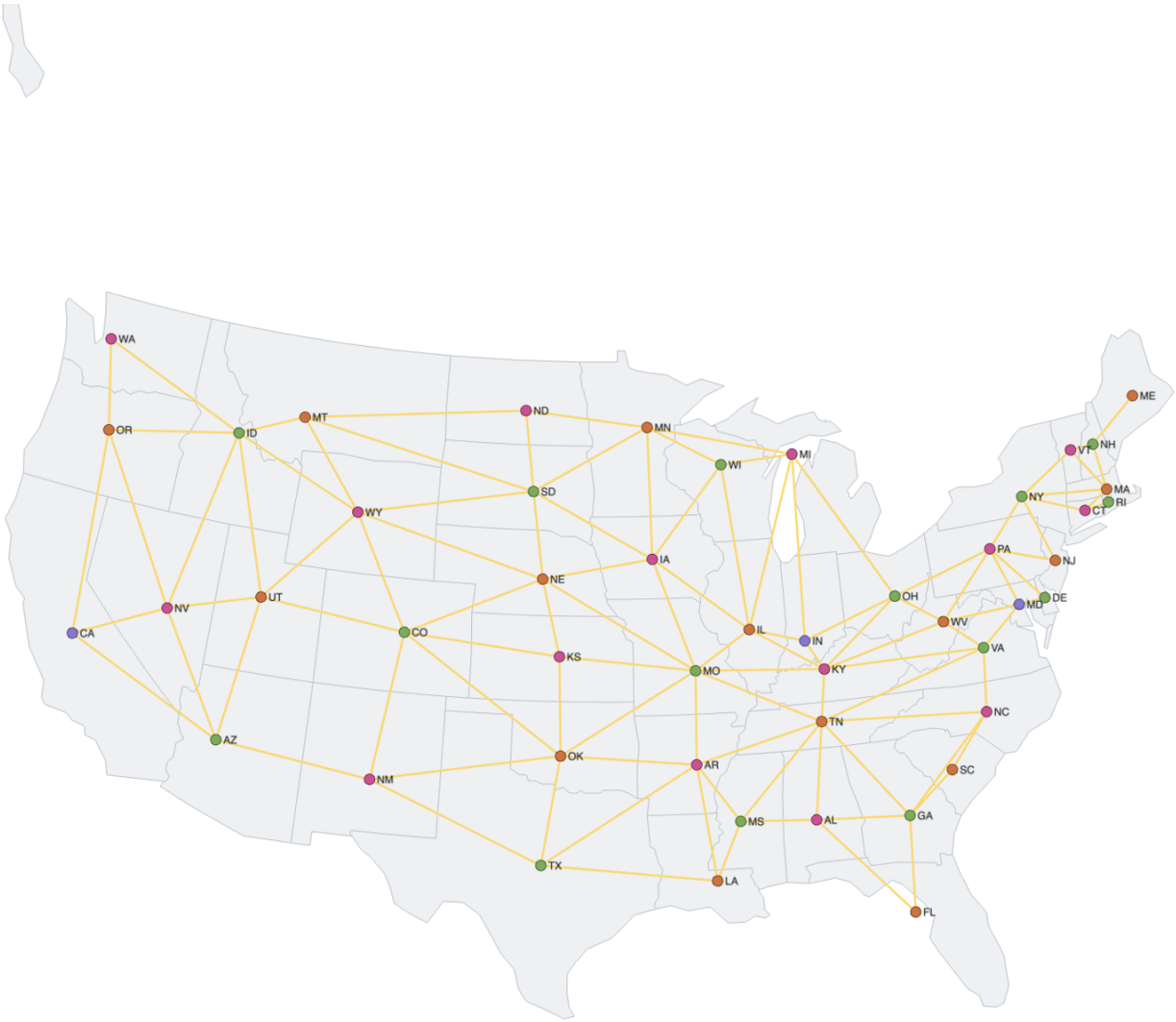


Figure 2: The lower 48 states, joined where they share a border, in four colours

How it works. `makeFromMap("usa")` makes a node for each of the lower 48 states and an edge wherever two share a border: 106 borders. Touching at a single point (the Four Corners) doesn't count. It places the states on a map (`geoLayout`) with their outlines behind. `makeFromMap("world")` does the same for 241 countries. `colorGraph()` gives each node a colour unlike its neighbours' (the DSATUR method), stores it in `color_class` and colours by it. A map needs at most four colours (the four-colour theorem), and the states take four. Region networks like these are *planar*: they can be drawn without crossing edges, and `edgeCrossings()` counts 0 here.

Variations. `colorGraph(strategy="largest_first")` uses a simpler greedy order and needs five

colours on the states: greedy colourings are not always optimal. `shortestPath(Washington, Florida)` counts border crossings between two states.

Pitfalls. The world map's drawing has 12 crossing edges: borders across the edges of the projection, and countries in several pieces, drawn at one point. Adjacency through the sea (ferries, tunnels) isn't a border here.

Draw flows, not lines Intermediate

You want the big corridors of a network to stand out from the clutter of individual edges.

```
makeFromGDF("us-airports.gdf")
subgraph((longitude > -130) & (latitude < 50), "lower 48")
geoLayout(projection="albers")
sizeBy(degree, 2, 20)
g.edges.color = "40,70,140,50"
bundleEdges()
```



Figure 3: Lower-48 routes, bundled into corridors

How it works. `subgraph` keeps the lower 48 states' airports and the routes among them, and running `geoLayout` again fits the map to them. `bundleEdges()` gathers edges that run alike into shared bundles, which turns a hairball of routes into a flow map: the transcontinental corridors, the north–south spines along each coast, and the fans around each hub (*Kitchen: Outline groups and bundle the hairball*). `method="fdeb"` gives smoother, more organic curves. `removeBundles()` goes back to straight lines.

Pitfalls. Bundling distorts geography on purpose: a bundled edge no longer shows the route it takes. Use it to show where flows concentrate, and draw individual routes straight when the route itself matters.

Hubs, bridges and weak points

Hubs are not the same as bridges Intermediate

```
makeFromGDF("us-airports.gdf")
for n in sortBy(degree)[-5:]: print(n.iata, n.city, n.degree,
 ↪  round(n.betweenness))
for n in sortBy(betweenness)[-5:]: print(n.iata, n.city, n.degree,
 ↪  round(n.betweenness))
```

How it works. The airports with the most routes (Atlanta, Chicago O'Hare, Denver, Dallas–Fort Worth, Minneapolis) are hubs. The airport on the most shortest paths is Anchorage, with only 34 routes. Most of Alaska's airports can reach the rest of the country only through it, so every journey between them and the lower 48 passes through one place. That's a bridge, and it's the more fragile kind of importance: other hubs can substitute for Atlanta, but nothing can substitute for Anchorage.

Do the hubs form a club? Intermediate

```
makeFromGDF("us-airports.gdf")
rc = richClub()
```

How it works. The *rich-club coefficient* for degree k is the density of the links among the nodes with more than k links. `richClub()` normalises it by the same measure in randomised networks with the same degrees: above 1, the hubs link to each other more than chance predicts. The US network is only weakly a rich club (1.05 to 1.10 for hubs with 25–75 routes): the big hubs are connected to each other, but no more than their sheer number of routes implies. Airline hub systems feed each hub from its own region rather than tying the hubs into a tight core.

Find the weak points Basic

```
makeFromGDF("us-airports.gdf")
geoLayout(projection="albers")
cuts = cutNodes()
cuts.color = red
cuts.size = 12
bridges().color = red
len(cuts), len(bridges())
```

How it works. A *cut node* is one whose loss disconnects part of the network (60 airports here), and a *bridge* is an edge whose loss does the same (165 routes). Most of them are small: a regional airport that is the only link for a still smaller one. A few, like Anchorage, hold up whole regions. Sort the cut nodes by how much they'd strand (next recipe) to tell them apart.

What if a hub closes? Intermediate

```
makeFromGDF("us-airports.gdf")
geoLayout(projection="albers")
```

```
ss("open")
remove(ANC)
colorize(component)
g.edges.color = "150,150,150,60"
len(giantComponent()), len(component.distinct)
```

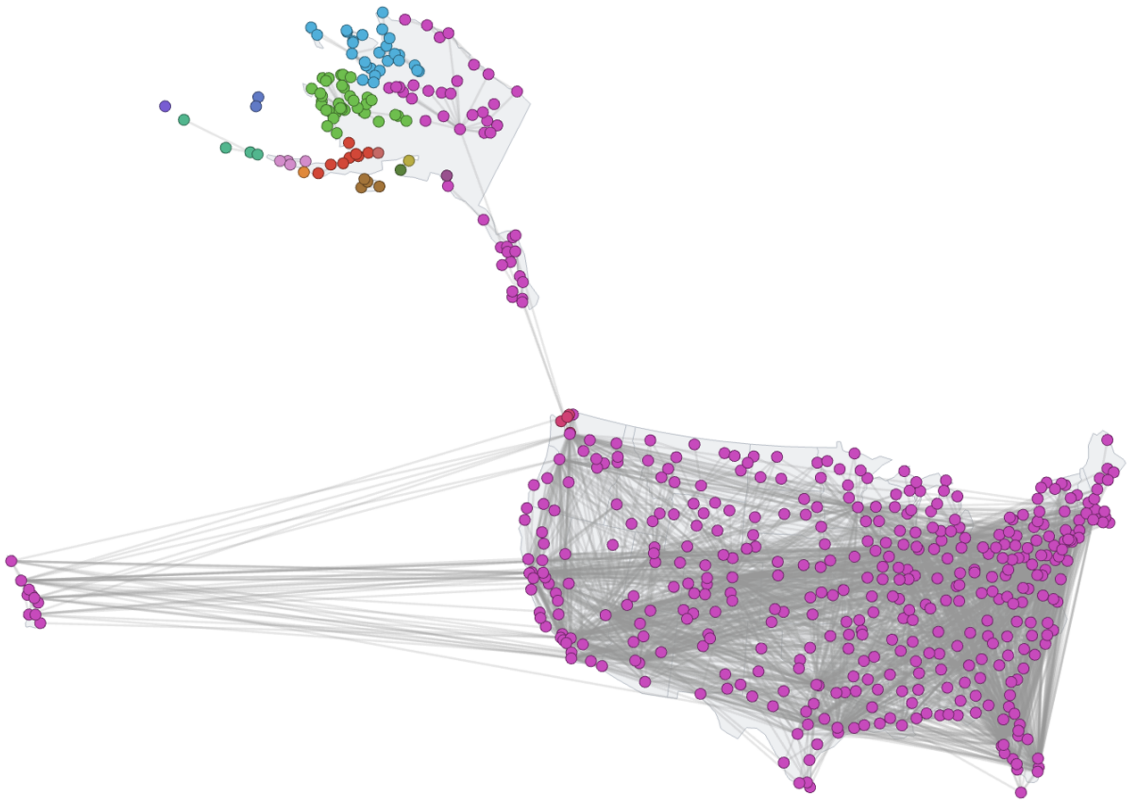


Figure 4: Without Anchorage, Alaska falls into pieces

How it works. `ss("open")` saves the network and `remove(ANC)` closes Anchorage. Computed fields such as `component` follow the change on their own: they're recalculated the next time they're read.

Colouring by component shows what's left. Eight airports were already off the main network (two small island systems), and closing Anchorage cuts off 87 more, in 12 new pieces, nearly all in Alaska. Closing Atlanta, the busiest hub, strands eleven. `ls("open")` reopens it, and `morph("open", 1500)` animates the difference.

Robustness

Failures versus attacks Intermediate

You want to know how the network degrades as parts fail: by accident (at random), or by targeted attack on its most important parts.

Ingredients: `us-airports.gdf` · `robustness`

```
makeFromGDF("us-airports.gdf")
r = robustness(["random", "degree", "betweenness", "adaptive"])
```

How it works. `robustness()` removes nodes in each given order and records the share of nodes still in the largest connected piece after each removal. It plots the curves and leaves the graph untouched. The orders are:

order	removes first	models
`"random"`	random nodes (averaged over `trials`)	failures, weather, accidents
`"degree"`	the nodes with the most links	an attack on the hubs
`"betweenness"`	the nodes on the most shortest paths	an attack on the bridges
`"adaptive"`	the current biggest hub, re-ranked after each removal	a determined attacker
a node field	the highest values of that field	any priority: age, cost, risk

The US airline network loses half its largest piece only after 34% of airports fail at random. An attack on the hubs does it with 6%, and an attack on the bridges with 3%. This *robust yet fragile* pattern is typical of networks with hubs. Random failures mostly hit small airports that nothing depends on, while a few removals in the right places split the system. The world network is similar (37% at random, 8% by degree).

The whole curve, fast. `percolation()` gives the same kind of curve in one pass, for every share of nodes at once: it adds the nodes back one at a time in random (or attack) order and merges the pieces they join (the Newman–Ziff algorithm), so it suits big networks and many trials. It returns (share present, largest piece) pairs and charts them:

```
makeFromFile("power-grid-region.gdf")
percolation(trials=20)                  # random failures
percolation("degree")                   # hubs removed first
percolation(bond=true, trials=20)       # lines fail instead of stations
```

A region of the Western US power grid (400 stations) loses half its largest piece when 27% of stations fail at random, but when only 6% are removed hubs first. The Internet sample of 300 networks of the 2006 Internet behaves the same way (41% against 14%), and so do the airports (37% against

6%). With `bond=true` lines rather than stations fail: the grid's largest piece falls below half the stations once about 40% of lines are down.

Pitfalls. Connectivity isn't capacity. A network can stay connected while its remaining hubs are overloaded, so simulate load (or use `maxFlow`) when capacity matters. In this data, "connected" also counts a weekly flight the same as a shuttle.

Routes

Fewest hops or shortest distance? Intermediate

```
makeFromGDF("us-airports.gdf")
geoDistance()
hops = shortestPath(BRW, MIA)
[n.iata for n in hops if n in g.nodes], round(sum(e.km for e in hops if e in
 ↪  g.edges))
for e in g.edges: e.weight = e.km
short = shortestPath(BRW, MIA, weighted=true)
[n.iata for n in short if n in g.nodes], round(sum(e.km for e in short if e in
 ↪  g.edges))
```

How it works. `geoDistance()` stores each route's great-circle length in kilometres, as the edge field `km`. `shortestPath` counts hops by default. With `weighted=true` it uses each edge's `weight` as a length, so copying `km` into `weight` gives the shortest route by distance. From Utqiaġvik (Barrow), Alaska, to Miami, both answers take three flights through Anchorage, but by fewest hops the route goes via Los Angeles (8,696 km), while the shortest by distance goes via Minneapolis (7,623 km). The longest domestic route in the data is Honolulu–New York JFK, 8,007 km.

Variations. `spanningTree()` gives the fewest edges that keep everything connected (with `weight` as cost, the cheapest network). `maxFlow(a, b)` and `minCut(a, b)` give how many separate routes join two places, and which few routes' loss would separate them.

A round trip through several cities Intermediate

```
makeFromGDF("us-airports.gdf")
geoLayout(projection="albers")
geoDistance()
trip = travelingSalesman([SEA, BOS, MIA, LAX, DEN], weight="km")
trip.color = red
[n.iata for n in trip if n in g.nodes], round(trip.length)
```

How it works. `travelingSalesman(stops, weight)` finds a short round trip through the stops (an approximation: the exact problem is intractable for many stops), travelling along existing routes with `km` as the length. It may pass through other airports. The result is the route's airports and flights in order, and `.length` is its total: here Seattle, Boston, Miami, Denver, Los Angeles and back, 11,705 km in five flights. `cycle=False` gives a one-way route that starts at the first stop.

Send a flow at least cost Advanced

```
makeFromGDF("us-airports.gdf")
geoDistance()
addEdgeField("capacity", DOUBLE, None)
for e in g.edges: e.capacity = e.weight
minCostFlow(SEA, MIA, 5, cost="km")
[(e.node1.iata, e.node2.iata, e.flow) for e in g.edges if e.flow > 0]
```

How it works. `minCostFlow(a, b, amount)` sends `amount` units from `a` to `b` as cheaply as possible, with each edge limited to its `capacity` and costing its `cost` field per unit. Here capacity is a stand-in, the number of airlines on each route, and cost is distance. The cheapest way to move five units from Seattle to Miami splits them between the direct route and a route through Tampa. That's the shape of real logistics problems: freight, evacuation, network bandwidth.

Regional systems

Find the regional systems Intermediate

```
makeFromGDF("us-airports.gdf")
geoLayout(projection="albers")
louvainClusters()
colorize(louvain_class)
for grp in groupBy(louvain_class)[:6]:
    top = sortBy(grp, degree)[-3:]
    print(len(grp), [n.iata for n in top])
```

How it works. Communities in a transport network are regions served mostly from within. Here they follow the airlines' hub systems more than the map: the East around Atlanta, Charlotte and Detroit; Texas around Dallas–Fort Worth; the Mountain West and the Pacific coast around Denver and Las Vegas; the Midwest around Chicago and Minneapolis; and two Alaskan systems, one around Anchorage and one in the southeast around Juneau. On the map the communities overlap, because a hub's reach is continental.

Case studies

Case study: Alaska's single point of failure Intermediate

```
makeFromGDF("us-airports.gdf")
alaska = (latitude > 51) & (longitude < -129)
len(alaska)
subgraph(alaska, "Alaska")
geoLayout(projection="albers", parallels=(55, 65), center=(-154, 60))
sizeBy(betweenness, 3, 30)
labelBy(iata)
```

Alaska has 135 airports with scheduled service, a quarter of the country's, and few roads between them, so the air network *is* the transport network. `subgraph` opens Alaska's airports and routes on their own, and `geoLayout` with Alaskan parallels draws them in proportion. Sized by betweenness, the structure is a hub and spokes of spokes: Anchorage at the centre, and regional hubs (Bethel, Fairbanks, Nome, Kotzebue, and Juneau in the southeast) each serving their own villages. `cutNodes()` finds 22 airports in Alaska whose loss would strand somebody, and Anchorage would strand nearly everyone. Back in the national network, `robustness("betweenness")` removes Anchorage first, which is why the bridge attack does so much damage so quickly.

Case study: hubs of the world Intermediate

```
makeFromGDF("world-airports.gdf")
geoLayout()
top_routes = sortBy(degree)[-10:]
top_bridges = sortBy(betweenness)[-10:]
[n.iata for n in top_routes], [n.iata for n in top_bridges]
(top_routes & top_bridges)
```

Six airports are on both top-ten lists (Amsterdam, Frankfurt, Paris, Istanbul, Chicago O'Hare and Beijing): the great hubs both have many routes and join regions. The other four on each list tell different stories. By routes they are Atlanta, Dallas–Fort Worth, Munich and Moscow Domodedovo, with many routes mostly within their own region. By betweenness they are Anchorage and Los Angeles across the Pacific, Dubai between Europe, Africa and Asia, and Toronto, which join continents with comparatively few routes. A planner asking "which airports matter most?" needs both lists: the first for traffic, the second for keeping regions connected at all.

Capstone

Capstone: an airport disruption simulator Advanced

The problem. An operations planner wants to rehearse disruptions. Close an airport (weather, a strike, an accident) and see who loses service. Close the top few by some measure and see how fast the system fragments. This recipe builds a simulator on the US network:

- the network on a map, airports sized by routes;
- a click closes an airport, and the airports cut off from the main network turn red;
- a drop-down and a slider close the top k airports by routes or by betweenness;
- a reset button reopens everything;
- the robustness curves sit beside the map for reference.

```
makeFromGDF("us-airports.gdf")
geoLayout(projection="albers")
sizeBy(degree, 2, 22)
g.edges.color = "130,130,130,35"
_ = robustness(["random", "degree", "betweenness"])
ss("open")
state = {"closed": []}
```

```
def update():
    main = giantComponent()
    g.nodes.color = "70,110,180"
    stranded = [n for n in g.nodes if n not in main]
    for n in stranded: n.color = red
    status.text = f"closed {len(state['closed'])}: {',
 ↪  '.join(state['closed'][:8])}; {len(stranded)} airports cut off"
    StatusBar.setStatus(status.text)

def close(n):
    state["closed"].append(n.iata)
    remove(n)
    update()

def reopen():
    ls("open")
    state["closed"] = []
    update()

def close_top(measure, k):
    reopen()
    ranked = sortBy(degree if measure == "routes" else betweenness)[-int(k):]
    for n in ranked:
        state["closed"].append(n.iata)
    remove(ranked)
    update()

pick = Dropdown(["routes", "betweenness"], label="close the top airports by")
count = Slider(0, 20, 3, label="how many", onChange=lambda v:
 ↪  close_top(pick.value or "routes", v))
status = Label("click an airport to close it")
ui.dock(Panel("Disruption", pick, count, Button("Reopen all", onClick=reopen),
 ↪  status, position="bottom"))
graphevents.clickNode = close
update()
center()
```

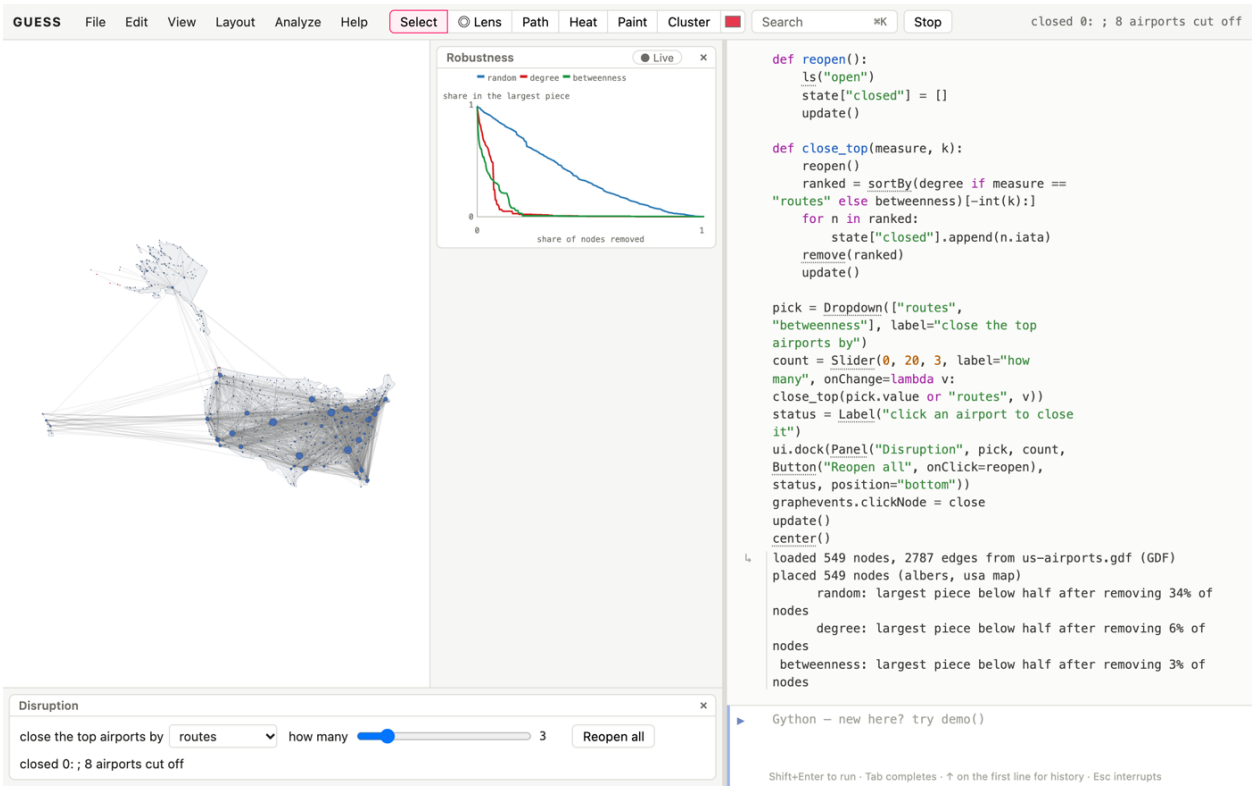


Figure 5: The disruption simulator: closing airports and seeing who is cut off

How it works.

- **Saving the network.** `ss("open")` saves the full network once. `reopen()` returns to it with `ls`.
- **Closing airports.** A click removes that airport. The slider ranks airports by routes (`degree`) or `betweenness`, and removes the top k.
- **Who's cut off.** After each change, `update()` finds the largest connected piece (as in the hub-closure recipe) and colours everyone outside it red. With the slider on betweenness, three closures are enough to strand most of Alaska.
- **The chart.** The robustness chart beside the map is the same analysis as the failures-versus-attacks recipe, for every k at once.

As an app. A capstone like this can be packaged so it opens on its own, without the console, through `?app=` (manual, *Apps*). *Help* → *Example apps* → *Airport Disruption* is one for this book: click an airport, close it, and see which airports are cut off from the rest. Its source is `web/public/apps/airports/`.